On the interpretation of the material conditional and the intuitive formalisation of implication by logical consequence

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Abstract

The material conditional $a \to b \equiv \neg a \vee b$ is a basic definition in formal logic and commonly used in mathematics. Minimalistic or sloppy translation to natural language has led to substantial confusion which can be restored either by making the translation precise or by a change in the logical formalism. Both involve leaving the result open in case the premise a is false. Separating this *irrelevant* case recovers a consistent formalisation of logical consequence *"If a, then b."* at the cost of ambiguity. Irrelevance is one way to introduce a paraconsistent logic and is realised in particular in the *Logic of Paradox* and *Relevance Logic* (see *G. Priest. An Introduction to Non-Classical Logic: From If to Is.* Cambridge, 2008 [1]). The discussed definition of implication breaks the principle of absolute logical equivalence by making it dependent on the truth value of a statement. For the case of an implication being true, logical equivalence and proofs remain classical, including indirect proofs. If an implication is false, differences appear due to ambiguity by irrelevant cases.

Material consequence and its interpretation

In mathematics the classical material consequence or conditional

$$a \to b \equiv \neg a \vee b \tag{1}$$

is often spoken as *"If a, then b."*, *"a implies b."*, *"b follows from a."* or *"a only if b."* which, as we will see, can fail to capture its actual meaning. Values of $a \to b$ and its descendents depending on truth values of a and b are shown in Table 1.

a	b	$a \to b$	$\neg(a \to b)$	$a \leftrightarrow b$
F	F	T	F	T
F	T	T	F	F
T	F	F	T	F
T	T	T	F	T

Table 1: Classical truth table of material consequence, its negation, and equivalence.

An intuitive interpretation of the material conditional translated to words runs into trouble for the case $a = F$ leading to $(a \to b) = T$, independent from b. A well-known feature of (ab-)using the material conditional as a rule of implication is the principle of explosion, i.e. a single false statement in a formal system can be used to prove arbitrary statements, thus destroying the system. This will be resolved at the end of the section.

Despite its counterintuitiveness, an advantage of setting $(a \to b) = T$ for $A = F$ lies in the simplification of proof. To show that $(a \to b) = T$ holds, it is enough to show that $a = T$ and $b = F$ cannot be realised at the same time. Therefore, to show the universal truth of $a \to b$, it is enough to show it under the condition that $a = T$, as it is true per definition for $a = F$. This also eases proofs of equivalence

$$a \leftrightarrow b \equiv (a \to b) \wedge (b \to a). \tag{2}$$

In order to show $(a \leftrightarrow b) = T$ it is enough to show truth of material consequences in both directions, thereby also covering the case $(a, b) = (F, F)$ per default.

This seemingly advantage turns into a disadvantage for the negation of the material consequence,

$$\neg(a \to b) \equiv a \wedge \neg b, \tag{3}$$

which would translate to *"If a, then not b"* when looking only at the last two columns of Table 1. To prove $\neg(a \to b) = T$ (i.e. $(a \to b) = F$) we need not only exclude the possibility $(a, b) = (T, T)$, but also avoid $a = F$, so prove that $a = T$. The latter refers to a alone and should intuitively have nothing to do with a relation between a and b.

The mentioned issues are resolved if language is made precise. A minimalistic way to talk about the material conditional would be directly based on its definition[1],

$$a \to b \equiv \neg a \vee b : \text{“}\textit{a is false, or b is true.}\text{”} \tag{4}$$

and for its negation

$$\neg(a \to b) \equiv a \wedge \neg b : \text{“}\textit{a is true, and b is false.}\text{”} \tag{5}$$

Here we see that in particular the negation of the material conditional intuitively *assigns* truth values to a and b. The statement $\neg(a \to b)$ doesn't describe a relation, but rather two facts.

For an interpretation of $a \to b$ as a consequence *“If a, then b”*, one could argue that the case $a = F$ is implicitly included by the law of the excluded middle, resulting in a true statement per default. As this can lead to misinterpretation, a slightly more redunant, but inuitive translation of the material consequence into words should be

$$a \to b : \text{“}\textit{If a, then b, or a is false.}\text{”} \tag{6}$$

and its negation[2]

$$\neg(a \to b) : \text{“}\textit{If a, then not b, and a is true.}\text{”} \tag{7}$$

Thus we could translate the meaning of the material conditional into words by making the extra conditions precise that we have omitted before.

The explosion problem is resolved at least intuitively, since the addition of *“or a is false”* when speaking of $a \to b$ in (6) exactly says that if a is false, so we don't care about the value of b. Only the omission of the last part of (7) leads to the incorrect interpretation that we could *conclude* on the truth of b from the material conditional. The material conditional says nothing about the truth value of b if $a = F$, but only of the overall expression $(a \to b) \equiv (\neg a \vee b) = T$. Explosion doesn't occur, as any chain of proofs containing a false premise in a material conditional says nothing about the truth of the conclusion. It is misleading and unintuitive to define a formal proof by checking if $a \to b$ is true, as one would intuitively believe to have proven b in this manner. The latter is only true for $a = T$, but not for $a = F$. This must be kept in mind when using “$\to$” as a rule of inference.

Logical consequence and irrelevance

The other possibility to fit intuitive language and logic consists in changing to a logical rule $\Rightarrow$ different from the material conditional $\to$ to represent the implication *“If a, then b.”* This can be done by starting with $\to$ and postulating a false premise $a = F$ doesn't matter. Introducing this ambiguity restores an intuitive definition of implication by the logical consequence[3] $a \Rightarrow b$. In such a case we say that the combinations (a, b) are *irrelevant* for $a \Rightarrow b$ and denote it with I rather than T or F. For logical primitives $\wedge$,$\vee$ involving I we introduce rules in Table 2 together with $\neg I = I$.

a	b	$a \vee b$	$a \wedge b$
F	I	I	F
T	I	T	I
I	F	I	F
I	T	T	I

Table 2: Logical primitives involving irrelevance.

The only cases *relevant* for $a \Rightarrow b$, where a fixed truth value should be assigned are the ones with $a = T$. After distinguishing irrelevant cases with $a = F$, the logical consequence $a \Rightarrow b$ can be written inside the truth table 3.

The material conditional and its negation can then be recovered by adding truth values for a via

$$a \to b \equiv (a \Rightarrow b) \vee \neg a, \tag{8}$$

$$\neg(a \to b) \equiv (a \Rightarrow b) \vee \neg a, \tag{9}$$

This also corresponds exactly to the verbose way of translating the material conditional into words in (6-7).

[1] Here *“a is false”* is intentionally used instead of *“not a”* and *“b is true”* instead of *“b”* alone, as the truth values are respectively identical, but the language is more clear in the present version.

[2] Of course the following sentence is a rather complicated way to describe (5), but it is useful, as it allows its last part to be omitted.

[3] often spoken as “*a entails b.*” and denoted by $\models$ in literature, but we intentionally keep the symbol $\Rightarrow$ which is often used with a logical, rather than a material consequence in mind.

a	b	$a \Rightarrow b$	$\neg(a \Rightarrow b)$	$a \Leftrightarrow b$	$\neg b \Rightarrow \neg a$
F	F	I	I	I	T
F	T	I	I	F	I
T	F	F	T	F	F
T	T	T	F	T	I

Table 3: Logical implications with irrelevance.

Weak equivalence

$$a \Leftrightarrow b \equiv (a \Rightarrow b) \wedge (b \Rightarrow a) \tag{10}$$

can be considered a mixture of $a \wedge b$ and $a \rightarrow b$ with only $a = T$ or $b = T$ being relevant, but not $(a, b) = (F, F)$. This reflects the fact that a proof from either direction by a logical consequence takes at least one to be true, and we didn't assign a fixed truth value for the case $(a, b) = (F, F)$. For strong equivalence ($a \leftrightarrow b$) we keep the classical definition (2) leading to values of Table 1. It is easily re-written in terms of weak equivalence by

$$a \leftrightarrow b \equiv (a \Leftrightarrow b) \wedge (\neg a \Leftrightarrow \neg b). \tag{11}$$

Thus, to show *strong* equivalency $\leftrightarrow$ of two statements it is not enough to prove "$\Rightarrow$" from both directions. Remember that when constructing a proof for $a \leftrightarrow b$ from both directions of $\rightarrow$, any sloppiness in language could be masked by the "dirty trick" of setting the default value $a \rightarrow b = T$ in case $a = F$. If this is replaced by irrelevance, the verification of both, $a \Rightarrow b$ and $b \Rightarrow a$ only states that if either $a = T$ or $b = T$, no contradiction to $a \leftrightarrow b$ arises. The case $(a, b) = (F, F)$ is irrelevant to both directions, and stays so when they are combined.

Example

Let us analyse the unintuitive behavior of the material conditional and its resolution by an example,

a: *"The moon is (entirely) made of cheese."* and

b: "*My birthday is on date x.*"

Statement a is false[4], and b may be either true or false, depending on x. According to the usual translation of material conditional, the statement

"If the moon is made of cheese, then my birthday is on date x."

is true, independent from x. This sentence may be interpreted as paradoxical as soon as you see me celebrating every day because we started with a false premise. The two possible resolutions are

1. The correct translation of the material conditional into words as
   "The moon is not made of cheese, or my birthday is on any date x you can imagine",
   or
   "If the moon is made of cheese, then my birthday is on date x, or the moon is not made of cheese",
   which are perfectly true statements.

2. The introduction of irrelevant cases for a false premise, in which case
   "If the moon is made of cheese, then my birthday is on date x"
   becomes an irrelevant statement which can be either true or false.

It should be noted that if we interpret

"If the moon is made of cheese, then I'll eat my hat (, or the moon is not made of cheese)."

in the sense of a material conditional including the part in brackets, I am actually saying

"The moon is not made of cheese, or I'll eat my hat".

In the material sense this statement is true, in the logical sense it is irrelevant without brackets, and true with brackets. Both variants leave me free choice whether or not to eat my hat, since *"the moon is made of cheese"* is false.

Representing statements by logical rules

For a more precise definition of logical equivalence and formal proof of statements in a system including irrelevance we will now split the system into cases where logical rules apply. This also helps to illustrate how equivalence between two statements becomes dependent on their truth value. An insightful way to view the concept of irrelevance is by separation of tables of allowed and forbidden combinations for the cases of a statement being true or false, respectively. The case $(a \Rightarrow b) = T$ is shown in Table 4.

Rules for $(a \Rightarrow b) = T$ are the same as for $(a \rightarrow b) = T$, leading to their equivalence in that case. Combinations with false premise $a = F$ are

[4] as in *"It has been checked that at least parts of the moon are rocks, and we know rocks are not cheese."*

allowed

a	b
F	F
F	T
T	T

forbidden

a	b
T	F

Table 4: Rules for $(a \Rightarrow b) = T \equiv (a \rightarrow b) = T \equiv (\neg b \rightarrow \neg a) = T$.

allowed for $(a \Rightarrow b) = T$, as this case is irrelevant for the truth of $a \Rightarrow b$, and are true per definition for $(a \rightarrow b) = T$. For a true premise $a = T$, only the case of a true consequence $b = T$ is allowed, but not $b = F$. From the perspective of rules of allowed and forbidden combinations of a and b, the statement $(a \Rightarrow b) = T$ is further equivalent the contrapositive $(\neg b \Rightarrow \neg a) = T$.

The negation $(a \Rightarrow b) = F$ $(\equiv \neg(a \Rightarrow b) = T)$ has rules listed in Table 5. Here the two additional irrelevant cases with $a = F$ appear in the allowed cases, as opposed to $(a \rightarrow b) = F$ with rules in Table 6. Allowing irrelevant combinations of a and b for $(a \Rightarrow b) = F$ has the advantage that for a false premise a, no artificial limitations are imposed, e.g.

"If the moon is made of cheese, then my birthday is not on day x."

can be true or not, independent from x.

allowed

a	b
F	F
F	T
T	F

forbidden

a	b
T	T

Table 5: Rules for $(a \Rightarrow b) = F$.

allowed

a	b
T	F

forbidden

a	b
F	F
F	T
T	T

Table 6: Rules for $(a \rightarrow b) = F$.

The case $(a \Rightarrow b) = F$ is clearly not equivalent to $(a \rightarrow b) = F$, where only a single case is allowed, and three are forbidden and also not and also not to the negated contrapositive $(\neg b \Rightarrow \neg a) = F$ listed in Table 7. Interestingly, $(a \Rightarrow b) = F$ is equivalent to $(b \Rightarrow a) = F$ and $(a \Leftrightarrow b) = F$.

Truth of strong equivalence and weak equivalence has the same rules (Table 8), but their falsehood does not, since the case $(a, b) = (F, F)$ is not allowed in the strong form.

allowed

a	b
F	T
T	F
T	T

forbidden

a	b
F	F

Table 7: Rules for $(\neg b \Rightarrow \neg a) = F$.

allowed

a	b
F	F
T	T

forbidden

a	b
F	T
T	F

Table 8: Rules for $(a \Leftrightarrow b) = T \equiv (a \leftrightarrow b) = T$.

allowed

a	b
F	T
T	F

forbidden

a	b
F	F
T	T

Table 9: Rules for $(a \leftrightarrow b) = F$.

Additional properties and formulation of proofs

For both, classical operators such as $\rightarrow$ and non-classical $\Rightarrow$, allowed and forbidden cases are complete, i.e. all cases are covered, and complementary to each other, i.e. no case can ever be allowed and forbidden at once. This feature is essential for the common definition of proof which we state as

To prove a statement to be true, only the realisation of allowed cases may be possible.

or

To prove a statement to be true, the possibility of a forbidden case to be realised must be excluded.

To formulate proofs, one must show that a forbidden case is never realised.

So to show $(a \Rightarrow b) = T$ one needs to show the impossibility of the case $(a, b) = (T, F)$. Ways to do this include the classical ones:

1. Direct proof. Take $a = T$ and deduce that $b = T$.

2. Indirect proof by contrapositive. Take $b = F$ and deduce $a = F$.

3. Indirect proof by contradiction. Assume $(a \Rightarrow b) = F$ and deduce that $(a, b) = (T, F)$ is possible, thereby violating a forbidden case and proofing the opposite $(a \Rightarrow b) = T$.

To show the negation $(a \Rightarrow b) = F$, one can use its equivalent $(a \Rightarrow \neg b) = T$ and apply the rules above.

Truth of weak or strong equivalence $(a \Leftrightarrow b) = T \equiv (a \leftrightarrow b) = T$ is shown as usual by $(a \Rightarrow b) = T$ and $(b \Rightarrow a) = T$, thus excluding mixed truth values for a and b. To show the negation of strong equivalence, $(a \leftrightarrow b) = F$, one must show that $(a, b) = (F, F)$ is impossible in addition to $(a \Leftrightarrow b) = F \equiv ((a \Rightarrow b) \wedge (b \Rightarrow a)) = F$.

Classical operators additionally fulfil the requirement of complementarity between allowed cases for a statement and its negation, e.g. $(a \to b) = T$ and $(a \to b) = F$ must not share any allowed cases. This leads to a single classical truth table, containing only T and F, like in Table 1. In non-classical operators there is no restriction, e.g. $(a \Rightarrow b) = T$ and $(a \Rightarrow b) = F$ share some allowed cases. These are the irrelevant cases in the above definition and are marked by I in a truth tables like Table 3.

Conclusion

After pointing out possible misunderstandings of the material conditional $a \to b$ and seemingly paradox results, two ways of resolution have been identified. The first consists in adding an extra clause to *"If a, then b"* when speaking of a material conditional. The principle of explosion can be avoided by restricting the use of $a \to b$ as a law of inference to true premisses a. Another alternative is the use of a logical consequence $a \Rightarrow b$ instead of a material conditional, containing the term *irrelevant* in addition to *true* and *false* for the case of a false premise a. Considering allowed and forbidden cases for statements a and b individually for true and false logical consequence, identities and techniques of proofs could be found to be classical if $a \Rightarrow b = T$, and different from the classical case for $a \Rightarrow b = F$. This leads to the conclusion that for $a = T$ it is irrelevant, which technique to use to prove $a \Rightarrow b = T$. For equivalance and negation, special care must be taken, especially when using the contrapositive. The case where irrelevant inputs a, b enter $a \Rightarrow b$ has not been discussed. This would most likely lead to a fourth possibility "unknown" or *"U"* to cover undecidable combinations.

References

[1] Graham Priest. *An Introduction to Non-Classical Logic: From If to Is.* Cambridge, 2008.